

CO4 AT2 – Coding Golf Competition

CSA 0315 – Data Structures

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1) Write a program to sort the array 50, 23, 9, 18, 61, 32 using Quick Sort.

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Q2
Bubble Sort
25 marks

Q3
Insertion Sort
25 marks

Q4
Merge Sort
25 marks

4/4 answered

Your Code

```
1 #include<stdio.h>
2 void quicksort(int a[50],int low,int high)
3 {
4     int i,j,p,t;
5     if(low<high)
6     {
7         p=a[low];
8         i=low;
9         j=high;
10        while(i<j)
11        {
12            while(a[i]<=p&& i<high)
13                i++;
14            while(a[j]>p)
15                j--;
16            if(i<j)
17            {
18                t=a[i];
19                a[i]=a[j];
20                a[j]=t;
```

Input

6
50 23 9 18 61 32

Output

9 18 23 32 50 61

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```
21     }
22 }
23 t=a[low];
24 a[low]=a[j];
25 a[j]=t;
26 quicksort(a,low,j-1);
27 quicksort(a,j+1,high);
28 }
29 }
30 int main( )
31 {
32     int i,n,a[50];
33     scanf("%d",&n);
34     for(i=0;i<n;i++)
35         scanf("%d",&a[i]);
36     quicksort(a,0,n-1);
37     for(i=0;i<n;i++)
38         printf("%d ",a[i]);
39     return 0;
40 }
```

Submitted

2) Write a program that implements the bubble sort

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Q2
Bubble Sort
25 marks

Q3
Insertion Sort
25 marks

Q4
Merge Sort
25 marks

4/4 answered

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[50],i,j,n,t;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=0;i<n-1;i++)
9         for(j=0;j<n-1-i;j++)
10            if(a[j]>a[j+1])
11            {
12                t=a[j];
13                a[j]=a[j+1];
14                a[j+1]=t;
15            }
16     for(i=0;i<n;i++)
17         printf("%d ",a[i]);
18     return 0;
19 }
```

Input
5
9 6 3 8 1

Output
1 3 6 8 9

3) Write a program sort the elements 9, 5, 1, 4, 3 using Insertion Sort.

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Q2
Bubble Sort
25 marks

Q3
Insertion Sort
25 marks

Q4
Merge Sort
25 marks

4/4 answered

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[50],i,j,n,t;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=1;i<n;i++)
9     {
10         j=i;
11         while(j>0&& a[j]<a[j-1])
12         {
13             t=a[j];
14             a[j]=a[j-1];
15             a[j-1]=t;
16             j--;
17         }
18     }
19     for(i=0;i<n;i++)
20         printf("%d ",a[i]);
21     return 0;
22 }
```

Input
5
9 5 1 4 3

Output
1 3 4 5 9

4) Apply Merge Sort to the array 38, 27, 43, 3, 9, 82, 10.



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Q2
Bubble Sort
25 marks

Q3
Insertion Sort
25 marks

Q4
Merge Sort
25 marks

4/4 answered

Your Code

```
1 #include<stdio.h>
2 void merge(int a[],int low,int mid,int high)
3 {
4     int t[20],i,j,k;
5     i=low;
6     j=mid+1;
7     k=low;
8     while(i<=mid&& j<=high)
9     {
10         if(a[i]<a[j])
11         {
12             t[k]=a[i];
13             i++;
14         }
15         else
16         {
17             t[k]=a[j];
18             j++;
19         }
20         k++;
21     }
```

Input

7
38 27 43 3 9 82 10

Output

3 9 10 27 38 43 82



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```
21     }
22     while(i<=mid)
23     {
24         t[k]=a[i];
25         i++;
26         k++;
27     }
28     while(j<=high)
29     {
30         t[k]=a[j];
31         j++;
32         k++;
33     }
34     for(i=low;i<=high;i++)
35         a[i]=t[i];
36 }
37 void mergesort(int a[],int low,int high)
38 {
39     int mid;
40     if(low<high)
41     {
42         mid=(low+high)/2;
```

```
41 {  
42     mid=(low+high)/2;  
43     mergesort(a,low,mid);  
44     mergesort(a,mid+1,high);  
45     merge(a,low,mid,high);  
46 }  
47 }  
48 int main()  
49 {  
50     int i,n,a[50];  
51     scanf("%d",&n);  
52     for(i=0;i<n;i++)  
53         scanf("%d",&a[i]);  
54     mergesort(a,0,n-1);  
55     for(i=0;i<n;i++)  
56         printf("%d ",a[i]);  
57     return 0;  
58 }
```

 Submitted